chapter progresses, we may add complexity or nuances to the starting problem to go deeper into some of the principles. For example, in chapter 2 we will start with the problem of maximizing rewards at a casino slot machine, and by solving that problem we'll cover most of the foundations of RL. Later we'll add some complexity to that problem and change the setting from a casino to a business that needs to maximize advertising clicks, which will allow us to round out a few more core concepts.

Although this book is for those who already have experience with the basics of deep learning, we expect to not only teach you fun and useful RL techniques but also to hone your deep learning skills. In order to solve some of the more challenging projects, we'll need to employ some of the latest advances in deep learning, such as generative adversarial networks, evolutionary methods, meta-learning, and transfer learning. Again, this is all in line with our skills-focused mode of teaching, so the particulars of these advances is not what's important.

Summary

- Reinforcement learning is a subclass of machine learning. RL algorithms learn by maximizing rewards in some environment, and they're useful when a problem involves making decisions or taking actions. RL algorithms can, in principle, employ any statistical learning model, but it has become increasingly popular and effective to use deep neural networks.
- The agent is the focus of any RL problem. It is the part of the RL algorithm that processes input to determine which action to take. In this book we are primarily focused on agents implemented as deep neural networks.
- The environment is the potentially dynamic conditions in which the agent operates. More generally, the environment is whatever process generates the input data for the agent. For example, we might have an agent flying a plane in a flight simulator, so the simulator would be the environment.
- The state is a snapshot of the environment that the agent has access to and uses to make decisions. The environment is often a set of